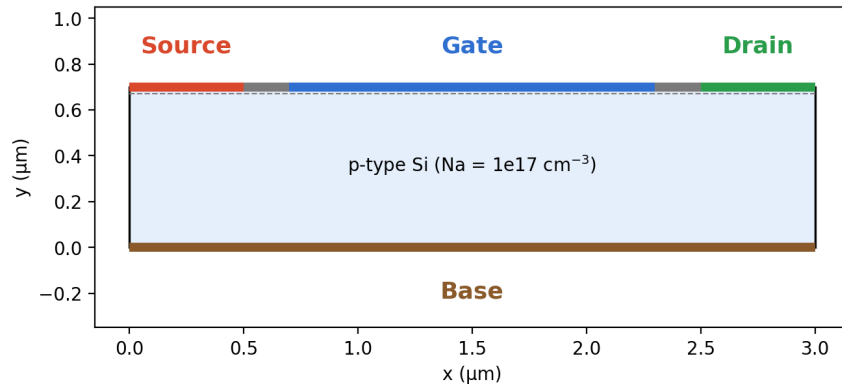


DC Characteristics of an n-Channel MOS Transistor (MOSFET)

Two-Dimensional Device Simulation Using the Semiconductor Module of COMSOL Multiphysics® 6.2



Simulated device cross-section: source, gate, drain and base contacts on a p-type silicon substrate

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Abstract

This project investigates the direct-current (DC) behaviour of an n-channel metal-oxide-semiconductor field-effect transistor (MOSFET) using two-dimensional numerical device simulation in COMSOL Multiphysics® 6.2 with the Semiconductor Module. The device is a $3\text{ }\mu\text{m} \times 0.7\text{ }\mu\text{m}$ p-type silicon structure (background acceptor concentration $N_a = 10^{17}\text{ cm}^{-3}$) with Gaussian-profiled n^+ source and drain implants (peak donor concentration $N_d = 10^{20}\text{ cm}^{-3}$), a 30 nm gate insulator of relative permittivity 4.5, and grounded source and substrate contacts. Fermi–Dirac statistics, Jain–Roulston band-gap narrowing and Shockley–Read–Hall recombination were included.

Two stationary studies were performed. In the first, the drain voltage was fixed at 10 mV and the gate voltage was swept from 0 to 4 V, giving a **threshold voltage of approximately 1.2 V**, in close agreement with the analytical estimate of 1.18 V. In the second, the drain voltage was swept from 0 to 5 V at gate voltages of 2, 3 and 4 V, reproducing the **linear, non-linear and saturation regions** of the output characteristic. The saturation current rises from about 25 μA at $V_G = 2\text{ V}$ to about 140 μA at 3 V and about 350–370 μA at 4 V, consistent with square-law behaviour, and a slight positive slope in saturation reveals the onset of short-channel effects. Electron-concentration and potential maps at $V_D = 0, 1$ and 5 V visualise the inversion channel and its **pinch-off** near the drain. Small-signal parameters, including output conductance and output resistance, were extracted from the simulated curves.

Keywords: MOSFET, COMSOL Multiphysics, semiconductor device simulation, threshold voltage, output characteristics, pinch-off, short-channel effect.

1. Introduction

1.1 Background

The MOSFET is by far the most widely used semiconductor device and is the primary building block of processors, memories and digital integrated circuits. Since the first microprocessors appeared roughly four decades ago, the device has undergone tremendous development, and it is now manufactured with feature sizes of 22 nm and below.

A MOSFET is essentially a miniaturised switch. In the structure studied here, the source and drain are ohmic contacts to heavily doped n-type regions formed in a p-type substrate. The gate electrode lies above the p-type region, slightly overlapping the two n-type regions, and is separated from the semiconductor by a thin insulating layer so that the gate and the underlying semiconductor form a capacitor. Applying a voltage to the gate alters the local band structure beneath it through the field effect. A sufficiently high gate voltage inverts a thin layer of the semiconductor from p-type to n-type; this inversion layer, or **channel**, connects the source and drain regions and has a much lower resistance than the back-to-back n^+p / pn^+ junctions that isolate them when the gate voltage is low. The gate voltage at which significant current begins to flow is called the **threshold** or turn-on voltage.

Numerical simulation is invaluable for studying such devices because it exposes internal quantities (carrier densities, potential, fields) that cannot be measured directly, and allows geometry, doping and bias to be varied rapidly and at no fabrication cost.

1.2 Objectives

- To build a two-dimensional physical model of an n-channel MOSFET in COMSOL Multiphysics® 6.2, including geometry, doping, contacts, gate insulator and mesh.
- To obtain the transfer characteristic (I_D versus V_G) at a small drain bias and to determine the threshold voltage, comparing it with an analytical estimate.
- To obtain the output characteristics (I_D versus V_D) for several gate voltages and identify the linear, non-linear and saturation regions.
- To visualise the electron concentration and electric potential inside the device and to explain current saturation through the pinch-off mechanism.
- To extract small-signal quantities (output conductance and resistance) from the simulated characteristics.

1.3 Scope of the Work

The study is restricted to DC (stationary) analysis of a single long-to-moderate channel device (gate length 1.6 μm) in two dimensions. Transient, small-signal AC, temperature-dependent and reliability effects are outside the scope of this work. The simulation follows the structure and procedure of the COMSOL Application Library model *DC Characteristics of a MOS Transistor*, with modifications to the geometry parameters and mesh settings described in Section 3.

2. Theoretical Background

2.1 Operating Regions of a MOSFET

For a gate voltage above threshold, the drain current of an n-channel MOSFET passes through three regimes as the drain voltage is increased:

- **Linear (ohmic) region** – at small V_D the channel behaves as a gate-controlled resistor and I_D increases almost linearly with V_D .
- **Non-linear region** – at intermediate V_D the potential drop along the channel reduces the local gate-to-channel voltage toward the drain, narrowing the channel and causing the I_D – V_D curve to bend over.
- **Saturation region** – when the channel is pinched off at the drain end, further increases in V_D produce little additional current, and the device behaves approximately as a voltage-controlled constant-current source.

Pinch-off occurs because the voltage between the gate and the channel varies with position: as current flows, the potential along the channel rises from the source toward the drain, so the inversion-layer width is no longer constant. A larger gate voltage yields a wider channel, lower resistance for a given drain voltage and a higher saturation current.

2.2 Threshold Voltage

The threshold voltage of a MOS structure may be approximated (Sze and Ng, 2007) by

$$V_T \approx V_{FB} + 2\psi_B + \frac{d_{ox} \sqrt{4\epsilon_{r,s}\epsilon_0 q N_a \psi_B}}{\epsilon_{r,ox} \epsilon_0} \quad (1)$$

where d_{ox} is the oxide thickness, $\epsilon_{r,ox}$ and $\epsilon_{r,s}$ are the relative permittivities of the oxide and semiconductor, ϵ_0 is the permittivity of free space, q is the elementary charge and N_a is the acceptor concentration under the gate. The flat-band voltage V_{FB} and the potential difference between the intrinsic level and the Fermi level, ψ_B , are

$$V_{FB} = \Phi_m - \chi + \frac{k_B T}{q} \ln \left(\frac{n_{eq}}{N_c} \right) \quad (2)$$

$$\psi_B = \frac{k_B T}{q} \ln \left(\frac{p_{eq}}{n_i} \right) \quad (3)$$

Here Φ_m is the work function of the metal contact, χ is the semiconductor electron affinity, k_B is Boltzmann's constant, T is the absolute temperature, N_c is the effective density of states in the conduction band and n_i is the intrinsic carrier density.

2.3 Equilibrium Carrier Densities

Assuming complete ionisation and Maxwell–Boltzmann statistics (reasonable under the gate), the equilibrium electron and hole densities are

$$n_{eq} = \frac{1}{2}(N_d - N_a) + \frac{1}{2}\sqrt{(N_d - N_a)^2 + 4n_i^2} \quad (4)$$

$$p_{eq} = -\frac{1}{2}(N_d - N_a) + \frac{1}{2}\sqrt{(N_d - N_a)^2 + 4n_i^2} \quad (5)$$

with N_d the donor concentration under the gate. For the substrate used here ($N_a = 10^{17} \text{ cm}^{-3}$, $N_d \approx 0$), these relations give $\psi_B \approx 0.42 \text{ V}$ for $n_i \approx 10^{10} \text{ cm}^{-3}$ at 300 K, so strong inversion begins when the surface potential reaches about $2\psi_B \approx 0.83 \text{ V}$. Together with the depletion-charge term of Eq. (1) ($\approx 1.25 \text{ V}$ for a 30 nm oxide of permittivity 4.5), this indicates that the threshold voltage is dominated by the oxide and doping parameters, while the flat-band voltage (which depends on the contact work function) shifts it by an additional negative offset. The reference model reports an analytical value of $V_T = 1.18 \text{ V}$.

2.4 Current–Voltage Relations and Small-Signal Parameters

For a long-channel device, the classical gradual-channel approximation gives, in the linear region and in saturation respectively,

$$I_{D,lin} \approx \mu_n C_{ox} \frac{W}{L} \left[(V_G - V_T) V_D - \frac{V_D^2}{2} \right] \quad (6)$$

$$I_{D,sat} \approx \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_G - V_T)^2 \quad (7)$$

$$V_{D,sat} \approx V_G - V_T \quad (8)$$

where μ_n is the electron mobility, C_{ox} the oxide capacitance per unit area and W/L the width-to-length ratio. Equation (7) predicts that the saturation current scales with the square of the gate overdrive ($V_G - V_T$), and Eq. (8) gives the saturation voltage. Real short-channel devices deviate from these expressions; nonetheless they remain a useful benchmark. The small-signal parameters of interest are

$$g_m = \frac{\partial I_D}{\partial V_G}, \quad g_d = \frac{\partial I_D}{\partial V_D}, \quad r_d = \frac{1}{g_d} \quad (9)$$

where g_m is the transconductance (slope of the transfer characteristic), g_d is the output conductance (slope of the output characteristic) and r_d is the output resistance. In an ideal long-channel device $g_d \rightarrow 0$ in saturation and $r_d \rightarrow \infty$; a finite r_d is a signature of channel-length modulation and other short-channel effects.

Internally, COMSOL solves the Poisson equation for the electrostatic potential together with the electron and hole continuity (drift–diffusion) equations, with the doping profile, recombination and boundary conditions supplying the source terms and constraints.

3. Model Definition and Simulation Methodology

This section documents the complete simulation set-up so that the study can be reproduced. The device is connected as shown in Fig. 1: both the source and the base (substrate) are grounded, while the drain and gate voltages are varied.

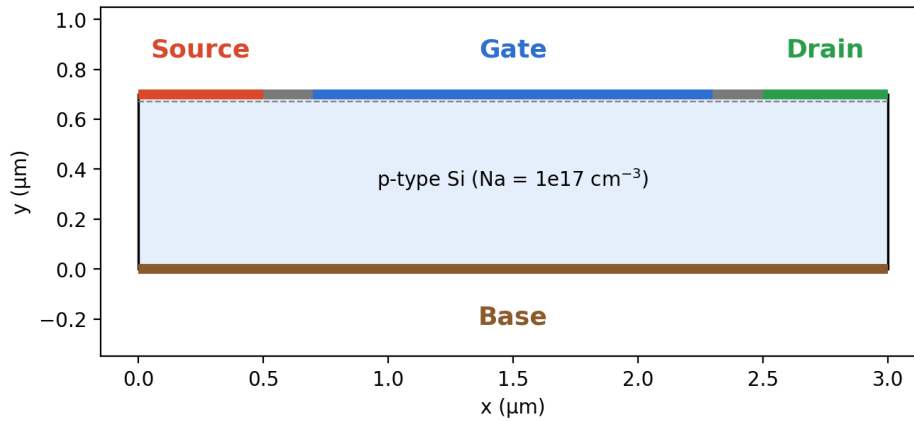


Figure 1: Model geometry with the source, gate, drain and base terminals and the p-type silicon substrate ($N_a = 10^{17} \text{ cm}^{-3}$).

3.1 Model Wizard and Global Parameters

A 2D model was created with the **Semiconductor (semi)** physics interface and a **Stationary** study, since a stationary solution is required to analyse the steady-state electric potential and current. Two global parameters were defined for later parametric sweeps:

Table 1: Global parameters

Name	Expression	Value	Description
V_d	10[mV]	0.01 V	Drain voltage
V_g	2[V]	2 V	Gate voltage

3.2 Geometry

The geometry was defined in micrometres. A rectangle of width 3 μm and height 0.7 μm represents the silicon substrate (a rectangular, two-dimensional cross-section of length L and thickness b). A closed-curve polygon was then added on top of the rectangle to define the split points for the source, gate and drain contacts, together with a line at $y = 0.67 \mu\text{m}$ that assists mesh generation. Its coordinates are given in Table 2.

Table 2: Polygon vertices (closed curve)

Vertex	1	2	3	4	5	6	7	8
x (μm)	0	0	0.5	0.7	2.3	2.5	3	3
y (μm)	0.67	0.7	0.7	0.7	0.7	0.7	0.7	0.67

Mesh Control Edges were used to control the size and distribution of the mesh by virtually removing selected geometric entities. Boundary numbering is shown in Fig. 2 and the role of each boundary is listed in Table 4.

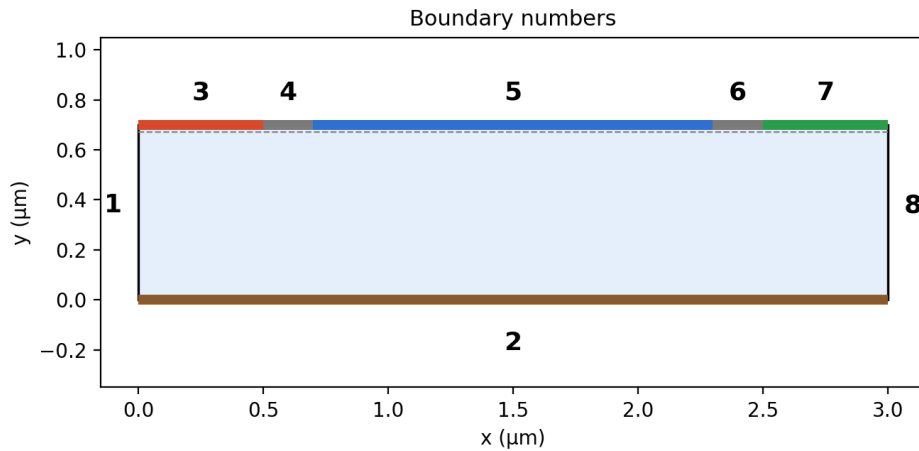


Figure 2: Boundary numbering of the geometry (boundaries 1–8).

3.3 Material and Physics Settings

Material. Silicon (Semiconductors > Si – Silicon) from the COMSOL material library was assigned to the domain. **Carrier statistics.** Fermi–Dirac statistics were selected in the Semiconductor interface, which remains accurate in the very heavily doped source and drain regions. **Band-gap narrowing.** The Jain–Roulston model was activated in the Semiconductor Material Model to account for the reduction of the band gap at high doping. **Recombination.** Trap-assisted (Shockley–Read–Hall) recombination with the default parameters was applied on all domains, so that carrier recombination and generation are considered throughout the geometry.

3.4 Doping Profiles

Three Analytic Doping Model features were used, all applied to all domains:

Table 3: Analytic doping models

Feature	Type / role	Key settings	Profile
Model 1	Uniform acceptor background	$N_{A0} = 1 \times 10^{17} \text{ cm}^{-3}$	Constant
Model 2	Source n^+ implant (box distribution)	$r_0 = (0, 0.6) \text{ } \mu\text{m}$; $W = 0.6 \text{ } \mu\text{m}$; $D = 0.1 \text{ } \mu\text{m}$; $N_{D0} = 1 \times 10^{20} \text{ cm}^{-3}$	Gaussian, $d_j = (0.2, 0.25) \text{ } \mu\text{m}$
Model 3	Drain n^+ implant (duplicate of Model 2)	$r_0 = (2.4, 0.6) \text{ } \mu\text{m}$; other settings as Model 2	Gaussian, $d_j = (0.2, 0.25) \text{ } \mu\text{m}$

For the box distribution, a rectangular region of constant doping is defined and the Gaussian fall-off occurs away from its edges. Because the implant is doped into a background of opposite type, the length scale d_j specifies the junction depth; different scales were used in the lateral ($x = 0.2 \text{ } \mu\text{m}$) and vertical ($y = 0.25 \text{ } \mu\text{m}$) directions. The background acceptor concentration was selected in the N_b list for the Gaussian profile.

3.5 Boundary Conditions

Table 4: Boundary assignments

Boundary	Location	Boundary condition	Setting
1, 8	Left / right walls	Insulation (default)	–
2	Bottom (base)	Metal contact, ideal ohmic	0 V (grounded)
3	$x = 0 - 0.5 \mu\text{m}$	Metal contact 1 (source), ideal ohmic	0 V (default terminal)
4, 6	Gaps between terminals	Insulation (default)	–
5	$x = 0.7 - 2.3 \mu\text{m}$	Thin insulator gate	$V_0 = V_g$; $\epsilon_{\text{ins}} = 4.5$; $d_{\text{ins}} = 30 \text{ nm}$
7	$x = 2.5 - 3 \mu\text{m}$	Metal contact 2 (drain), ideal ohmic	$V_0 = V_d$

The Metal Contact feature is a terminal-type boundary condition; by default it applies a fixed potential of 0 V, which is appropriate for the grounded source and base. The gate dielectric is not represented explicitly: the **Thin Insulator Gate** condition models both the gate contact and the thin oxide layer through its permittivity and thickness. Its charge setting refers to charge on the conductor and not to trapped charge at the oxide–semiconductor interface.

3.6 Mesh

A user-controlled mesh was used, combining an edge mesh along the contact boundaries (where the surface channel forms and gradients are steepest), a mapped mesh in the substrate with a graded boundary-layer distribution, and free triangular elements elsewhere. The maximum element growth rate was set to 1.05. Table 5 lists the settings, including those that were changed relative to the reference model.

Table 5: Mesh settings

Setting	Value used	Comment
Sequence type	User-controlled	–
Maximum element growth rate	1.05	Smooth size transition
Edge mesh: boundaries	3 – 7	Contact and gap boundaries
Edge mesh: max. element size	$0.08 \mu\text{m}$	Modified (reference: $0.03 \mu\text{m}$)
Mapped mesh: domain	Domain 2	Adjust edge mesh enabled
Distribution: elements	10	Modified (reference: 8)
Distribution: element ratio	10	Modified (reference: 9)
Growth / direction	Exponential, reversed	Fine near the surface

3.7 Studies

Study 1 – transfer characteristic. A stationary study with an auxiliary sweep (all combinations) was defined with $V_d = 0.01$ V fixed and $V_g = \text{range}(0, 0.2, 1.4), 2, 3, 4$ V. The gate voltage is swept with uneven steps to reduce computation time and file size; continuation on V_g uses the previous solution as the initial guess and lets the solver insert intermediate steps if necessary.

Study 2 – output characteristics. A second stationary study swept V_d from 0 to 5 V for $V_g = 2, 3$ and 4 V. Results were post-processed with 1D plot groups of the terminal current (semi.IO_2, in μA) and with 2D surface plots of the electron concentration and the electric potential.

4. Results and Discussion

4.1 Doping Distribution

Fig. 3 shows the donor concentration of the source implant (Analytic Doping Model 2): a plateau of 10^{20} cm^{-3} in the box region ($x \leq 0.6 \text{ } \mu\text{m}$, $y \geq 0.6 \text{ } \mu\text{m}$) with a Gaussian roll-off beyond its edges. Duplicating the feature at $x = 2.4 \text{ } \mu\text{m}$ produces the drain implant, and the signed dopant concentration $N_d - N_a$ (Fig. 4) shows both n^+ regions embedded in the p-type substrate.

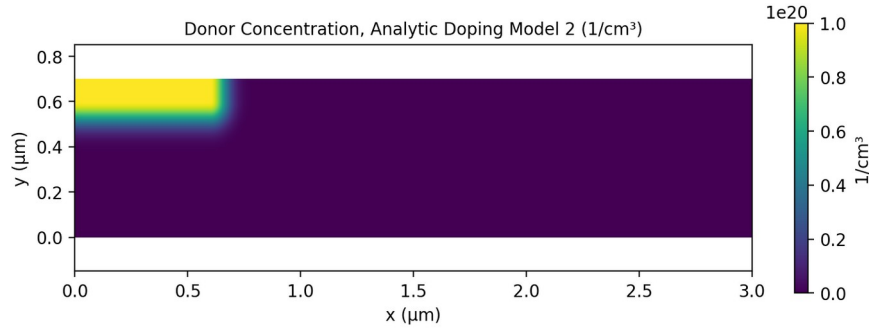


Figure 3: Donor concentration of Analytic Doping Model 2 (source implant).

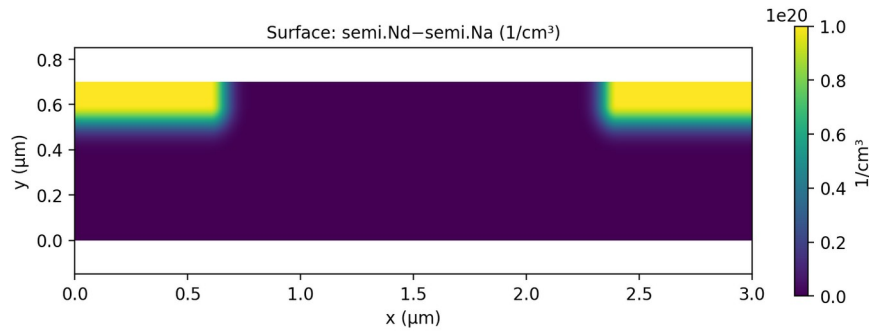


Figure 4: Signed dopant concentration $N_d - N_a$ showing the source and drain regions.

The net doping plot (Fig. 5) uses a red scale for p-type and a blue scale for n-type material on a logarithmic axis. The metallurgical junctions lie at a depth of roughly $0.35 \text{ } \mu\text{m}$ below the top surface ($y \approx 0.35 \text{ } \mu\text{m}$), and the lateral diffusion under the gate edges produces the required overlap between the n^+ regions and the gate. The n^+ regions extend to about $x \approx 0.8 \text{ } \mu\text{m}$ on the source side and start at about $x \approx 2.2 \text{ } \mu\text{m}$ on the drain side, so the metallurgical channel length is approximately $1.4 \text{ } \mu\text{m}$, slightly shorter than the $1.6 \text{ } \mu\text{m}$ gate.

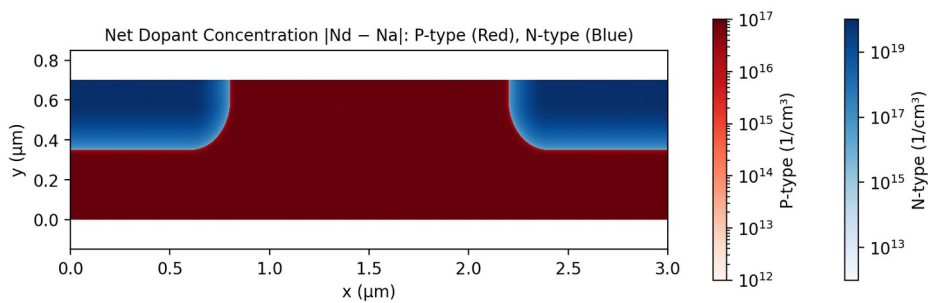


Figure 5: Net dopant concentration $|N_d - N_a|$ - p-type (red) and n-type (blue).

4.2 Transfer Characteristic and Threshold Voltage

Fig. 6 shows the drain current as a function of gate voltage for $V_D = 10$ mV. For V_G below about 1 V the current remains at its very low off-state level, then rises sharply as the inversion layer forms, and beyond about 1.5 V increases almost linearly with V_G , as expected for a device operating in the linear region with a small drain bias. The threshold voltage read from the curve is $V_T \approx 1.2$ V.

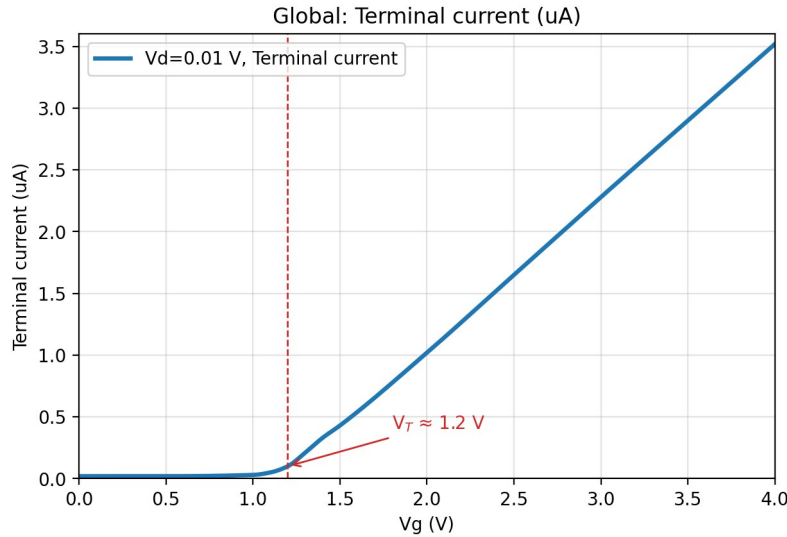
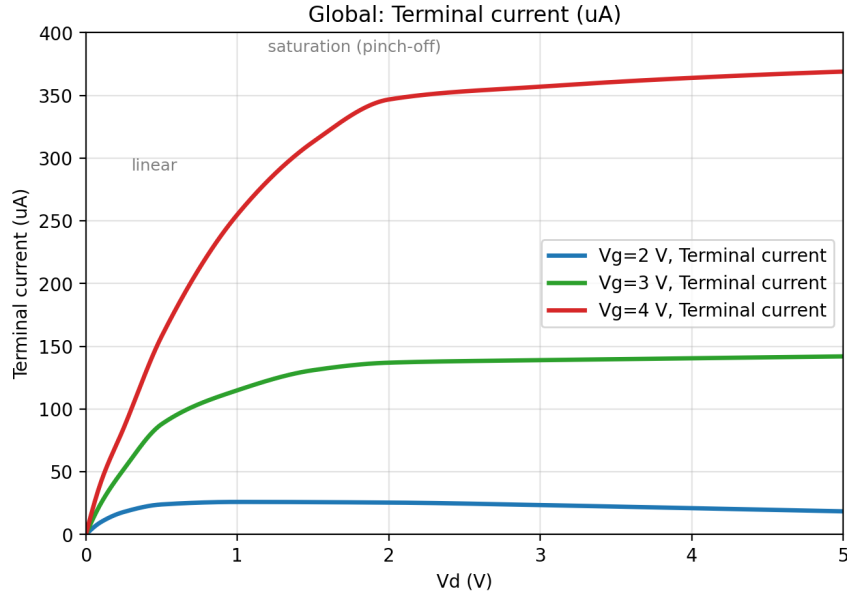


Figure 6: Drain current versus gate voltage for $V_D = 10$ mV; threshold voltage $V_T \approx 1.2$ V.

As a cross-check, extrapolating the linear portion of the curve (about $1.0 \mu\text{A}$ at $V_G = 2$ V and $3.5 \mu\text{A}$ at 4 V, giving a slope of $\approx 1.25 \mu\text{S}$) back to zero current yields an intercept of approximately 1.18 V, which agrees with the simulated value and with the analytical estimate of 1.18 V obtained from Eq. (1) in the reference model. The small discrepancy between analytical and numerical thresholds is expected, because the analytical expression assumes complete ionisation, Maxwell–Boltzmann statistics and a one-dimensional, long-channel structure. Between the off state ($\approx 0.02 \mu\text{A}$) and $V_G = 4$ V ($\approx 3.5 \mu\text{A}$) the current changes by more than two orders of magnitude at $V_D = 10$ mV, confirming that the gate effectively switches the device from a high-resistance to a low-resistance state.

4.3 Output Characteristics

Fig. 7 presents I_D versus V_D for $V_G = 2, 3$ and 4 V. All three curves display the classical three regions: a linear region at low V_D , a non-linear region at intermediate voltages, and an approximately constant-current saturation region at higher V_D . The current level increases strongly with gate voltage, since a larger V_G produces a wider, more conductive channel.

Figure 7: Drain current versus drain voltage for $V_G = 2, 3$ and 4 V.**Table 6: Output-characteristic summary (values read from Fig. 7)**

V_G (V)	Overdrive $V_G - V_T$ (V)	Approx. knee (saturation onset) V_D (V)	Saturation current $I_{D,sat}$ (μ A)	$K = 2I_{D,sat}/(V_G - V_T)^2$ (μ A/V ²)
2	0.8	≈ 0.5	≈ 25	≈ 78
3	1.8	≈ 2	$\approx 137 - 142$	≈ 85
4	2.8	≈ 2	$\approx 347 - 369$	≈ 89

The last column tests the square-law relation of Eq. (7): if the device followed it, $K = \mu_n C_{ox} W/L$ would be independent of V_G . The computed values fall within about $\pm 7\%$ of a mean of $\approx 84 \mu\text{A/V}^2$, confirming that the saturation current scales approximately with $(V_G - V_T)^2$. The remaining deviations are attributed to mobility degradation and short-channel effects.

Two features deserve comment. First, at $V_G = 3$ V the knee occurs near $V_D \approx 2$ V, close to the long-channel prediction $V_{D,sat} = V_G - V_T \approx 1.8$ V, whereas at $V_G = 4$ V the observed knee (≈ 2 V) is well below the predicted 2.8 V. Saturation therefore sets in earlier than the simple theory predicts at high overdrive, a typical signature of velocity saturation and short-channel behaviour. Second, the curves are not perfectly flat in saturation: at $V_G = 4$ V the current creeps from about $347 \mu\text{A}$ at 2 V to about $369 \mu\text{A}$ at 5 V. This slight gradient is due to the onset of short-channel effects, which mean that the standard analytic expressions for saturation voltage and current are inaccurate, although the predicted saturation voltages are of a similar magnitude.

At $V_G = 2$ V, which is only 0.8 V above threshold, the current is small ($\approx 25 \mu\text{A}$) and decreases slightly at large V_D (to $\approx 18 \mu\text{A}$ at 5 V). In this low-overdrive case the channel is very thin and strongly influenced by the drain field, and the gate control near the drain end is weakened as V_D grows; the decrease is thus a second-order effect of the short-channel geometry and the drain-induced potential redistribution rather than a failure of saturation.

4.4 Channel Formation and Pinch-off

Fig. 8 shows the electron concentration (left) and electric potential (right) at $V_G = 4$ V for $V_D = 5$ V, 1 V and 0 V. The electron concentration spans more than fifteen orders of magnitude, from about 10^4 cm⁻³ in the depleted substrate to about 10^{20} cm⁻³ in the n⁺ source and drain.

Electron Concentration (left) and Electric Potential (right)

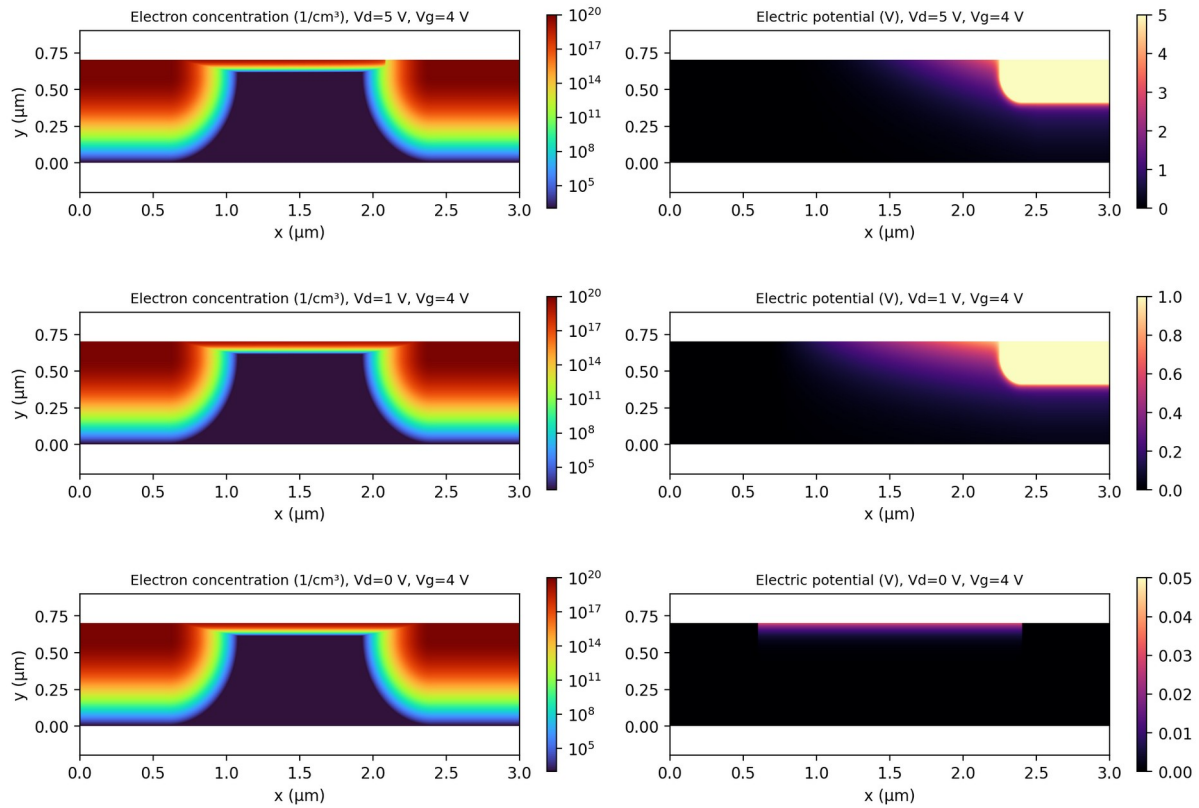


Figure 8: Electron concentration (left) and electric potential (right) at $V_G = 4$ V for $V_D = 5$ V (top), 1 V (middle) and 0 V (bottom).

- **$V_D = 0$ V (equilibrium in the channel).** A thin, uniform electron layer just below the gate oxide (of the order of tens of nanometres thick) connects the source and drain n⁺ regions: this is the inversion channel. The potential is essentially zero everywhere apart from small variations (< 0.05 V) beneath the gate, and no current flows because there is no lateral field.
- **$V_D = 1$ V.** The channel remains continuous, but a potential gradient now exists along it. The drain-side potential rises along the channel and the gate-to-channel voltage falls toward the drain; at this bias the thinning of the electron layer is still slight, corresponding to the non-linear part of the I–V curve.
- **$V_D = 5$ V.** The potential in the drain region is 5 V and the potential near the surface rises steeply towards the drain end of the channel. The gate-to-channel voltage is reduced there, so the inversion layer at the drain end is markedly thinner than at the source end. This is the **pinch-off** condition: the channel narrows at one end because of the field parallel to the surface, and the current saturates.

This picture explains why the drain current is nearly independent of V_D in saturation: once the channel is pinched off, additional drain voltage is dropped across the narrow pinched region between the channel end and the drain, while the voltage along the conducting channel stays

approximately fixed at $V_G - V_T$. Carriers reaching the pinched-off region are swept into the drain by the high field.

4.5 Small-Signal Parameters

The slope of the $V_G = 4$ V output characteristic between $V_D = 1$ V ($y_1 = 255.09$ μ A) and $V_D = 2$ V ($y_2 = 346.77$ μ A) was calculated from the simulated data:

$$g_d = \frac{y_2 - y_1}{x_2 - x_1} = \frac{346.77 - 255.09}{2 - 1} = 91.68 \mu\text{S} \quad (10)$$

$$r_d = \frac{1}{g_d} = \frac{1}{91.68 \mu\text{S}} \approx 10.9 \text{ k}\Omega \quad (11)$$

Because this is the slope of I_D with respect to V_D , it is the **output conductance** g_d and the corresponding output resistance r_d (≈ 0.0109 M Ω). It applies to the pre-saturation, non-linear part of the curve and is therefore relatively large. In true saturation (V_D from 2 V to 5 V) the same calculation gives a slope of only $(369 - 347)/3 \approx 7$ μ S, i.e. $r_d \approx 135$ k Ω – an order of magnitude higher, illustrating that the device approaches an ideal current source in saturation, with the finite r caused by short-channel effects.

The transconductance $g_m = \partial I_D / \partial V_G$ can be obtained from the transfer characteristic: at $V_D = 10$ mV, the linear region of Fig. 6 gives $g_m \approx 1.25$ μ S. From the output curves at $V_D = 5$ V, the current increases from about 142 μ A at $V_G = 3$ V to about 369 μ A at 4 V, giving a saturation transconductance $g_m \approx 227$ μ S. The corresponding intrinsic voltage gain g_m/g_d is therefore of the order of 30, demonstrating the amplifying capability of the device in saturation.

4.6 Summary of Results

Table 7: Key results

Quantity	Value	Reference / comment
Threshold voltage (simulation)	≈ 1.2 V	$V_D = 10$ mV
Threshold voltage (analytical)	1.18 V	Eq. (1), reference model
Saturation current, $V_G = 2$ V	≈ 25 μ A	Low overdrive
Saturation current, $V_G = 3$ V	$\approx 137 - 142$ μ A	–
Saturation current, $V_G = 4$ V	$\approx 347 - 369$ μ A	Slight slope: short-channel effect
Output conductance g_d ($V_D = 1 - 2$ V, $V_G = 4$ V)	91.68 μ S	$r_d \approx 10.9$ k Ω
Output conductance in saturation (2 – 5 V)	≈ 7 μ S	$r_d \approx 135$ k Ω
Saturation transconductance g_m ($V_D = 5$ V, $V_G = 3 - 4$ V)	≈ 227 μ S	Read from Fig. 7

5. Limitations and Sources of Error

- **Two-dimensional model.** The simulation is 2D, so the terminal current is expressed per unit out-of-plane depth as set in the model (the default was retained). Absolute current levels should be interpreted qualitatively and compared with other results only in relative terms.
- **Mesh resolution.** The maximum edge element size of 0.08 μm is coarser than the 0.03 μm of the reference model, and the boundary-layer distribution was altered. Small differences in the curves relative to the reference are therefore expected; a mesh-convergence study would quantify them.
- **Graph-derived values.** Several quantities in Tables 6 and 7 were read from the plots and are approximate (typically $\pm 3\text{--}5\%$).
- **Simplified gate stack.** The oxide is modelled through the Thin Insulator Gate condition, so interface traps, fixed oxide charge, quantum confinement and gate leakage are not included.
- **Physical models.** Default mobility and SRH parameters were used, no self-heating or temperature dependence was studied, and the analytical threshold comparison relies on complete ionisation and Maxwell–Boltzmann statistics.
- **Threshold definition.** The threshold voltage was determined graphically and by linear extrapolation; other definitions (constant-current, maximum transconductance) could give slightly different values.

6. Conclusions

A two-dimensional n-channel MOSFET was modelled in COMSOL Multiphysics® 6.2 and its DC characteristics were successfully simulated. The main conclusions are:

1. The threshold voltage of the device is approximately **1.2 V**, in good agreement with the analytical value of 1.18 V and with the extrapolated linear-region intercept.
2. The output characteristics exhibit linear, non-linear and saturation regions; the saturation current increases from about 25 μA to about 140 μA and about 350–370 μA as V_G is raised from 2 V to 3 V and 4 V, scaling approximately with the square of the gate overdrive ($K \approx 84 \mu\text{A}/\text{V}^2$).
3. Electron-concentration and potential maps confirm the formation of an inversion channel under the gate and its **pinch-off** at the drain end for high V_D , which is the physical origin of current saturation. In saturation the device behaves approximately as a constant-current source.
4. The slight rise of current in saturation and the earlier-than-predicted onset of saturation at high V_G show the influence of **short-channel effects**, which make the simple long-channel expressions only approximately valid.
5. The output conductance is about 92 μS ($r_d \approx 10.9 \text{ k}\Omega$) in the non-linear region and only about 7 μS ($r_d \approx 135 \text{ k}\Omega$) in saturation, and the intrinsic gain in saturation is of order 30.

Overall, the study demonstrates that finite-element device simulation reproduces the essential physics of MOSFET operation and provides insight into internal quantities that are inaccessible experimentally.

7. Future Scope

- Perform a mesh-convergence study and compare the 0.08 μm and 0.03 μm edge meshes quantitatively.
- Vary the gate length, oxide thickness, substrate doping and junction depth to study their influence on threshold voltage, saturation current and short-channel effects (drain-induced barrier lowering, channel-length modulation).
- Study the sub-threshold region on a logarithmic current scale to extract the sub-threshold swing and on/off ratio.
- Investigate other semiconductor materials (for example germanium or III–V channels) and high-k gate dielectrics.
- Add a small-signal (frequency-domain) analysis to determine capacitances and cut-off frequency, and a transient study for switching behaviour.
- Include temperature dependence and interface-trap models to assess reliability and variability.

References

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Appendix : Modelling Workflow Summary

The sequence of steps followed in COMSOL Multiphysics® 6.2 is summarised below.

Step	Stage	Main actions
1	Model Wizard	New > Model Wizard > 2D > Semiconductor (semi) > Study > Stationary
2	Global definitions	Parameters: $V_d = 10[\text{mV}]$, $V_g = 2[\text{V}]$
3	Geometry	Units μm ; rectangle 3×0.7 ; closed polygon (Table 2); mesh control edges; remove boundary 4 of the “fin”
4	Materials	Add Silicon to the component
5	Semiconductor	Fermi–Dirac statistics; Analytic Doping Models 1–3
6	Boundary conditions	Metal contacts (source B3, drain B7, base B2); Thin Insulator Gate B5 ($\epsilon = 4.5$, 30 nm)
7	Recombination / BGN	Trap-assisted recombination on all domains; Jain–Roulston band-gap narrowing
8	Mesh	User-controlled: edge mesh (0.08 μm), mapped mesh with distribution, free triangular
9	Plots	Signed dopant concentration (semi.Nd – semi.Na)
10	Study 1	$V_d = 0.01 \text{ V}$; $V_g = \text{range}(0, 0.2, 1.4), 2, 3, 4 \text{ V}$; plot semi.IO_2 ($I_D - V_G$)
11	Study 2	V_d sweep 0 – 5 V at $V_g = 2, 3, 4 \text{ V}$; plot semi.IO_2 ($I_D - V_D$)
12	Post-processing	Electron concentration and electric potential at $V_d = 5, 1, 0 \text{ V}$ ($V_g = 4 \text{ V}$)